

DAM WATER LEVEL ALERT SYSTEM USING ESP8266 AND HC-SR04

1. Problem Statement

Dams are important for water storage, irrigation, power generation, and flood control. During heavy rainfall, water levels can rise rapidly and may lead to overflow conditions. Manual monitoring of water levels is time-consuming and may not provide timely alerts. Therefore, an automated system is required to continuously monitor water levels and provide alerts when the water reaches critical levels.

The objective of this project is to develop an IoT-based Dam Water Level Alert System using an ESP8266 NodeMCU and HC-SR04 ultrasonic sensor. The system monitors water levels in real time, provides visual alerts through LEDs, and uploads data to the ThingSpeak cloud platform for remote monitoring.

2. Scope of Solution

The proposed system continuously measures the distance between the ultrasonic sensor and the water surface. Based on the measured distance, the water level percentage is calculated and classified into Safe, Warning, and Danger levels.

The system provides:

- Real-time water level monitoring.
- LED-based alert indication.
- Cloud data logging using ThingSpeak.
- Water level percentage calculation.
- Overflow alert indication.

3. Components Required

Hardware Components

- ESP8266 NodeMCU
- HC-SR04 Ultrasonic Sensor
- Green LED
- Red LED
- Two 2.2 k Ω Resistors

- Breadboard
- Jumper Wires
- USB Cable
- Water Bottle (Prototype Reservoir)

Software Components

- Visual Studio Code
- PlatformIO
- ThingSpeak Cloud Platform
- EasyEDA
- Fritzing

4. Circuit Description

The HC-SR04 ultrasonic sensor is connected to the ESP8266 NodeMCU. The TRIG and ECHO pins are connected to GPIO pins of the ESP8266. Two LEDs are connected through current-limiting resistors to indicate different water-level conditions.

Connections

- HC-SR04 VCC → 3.3V
- HC-SR04 GND → GND
- HC-SR04 TRIG → GPIO5 (D1)
- HC-SR04 ECHO → GPIO4 (D2)
- Green LED → GPIO0 (D3)
- Red LED → GPIO2 (D4)

5. Working Principle

- Warning Level: 40–80%
- Danger Level: 80–100%

LEDs indicate the current water level condition, and the data is uploaded to the ThingSpeak cloud platform for remote monitoring.

6. ThingSpeak Cloud The HC-SR04 ultrasonic sensor emits ultrasonic waves toward the water surface. The reflected waves are received back by the sensor.

Using the time taken for the echo to return, the ESP8266 calculates the distance between the sensor and the water surface.

The water level percentage is calculated from the measured distance and compared with predefined thresholds.

Alert Levels

- Safe Level: 0–40%

Integration

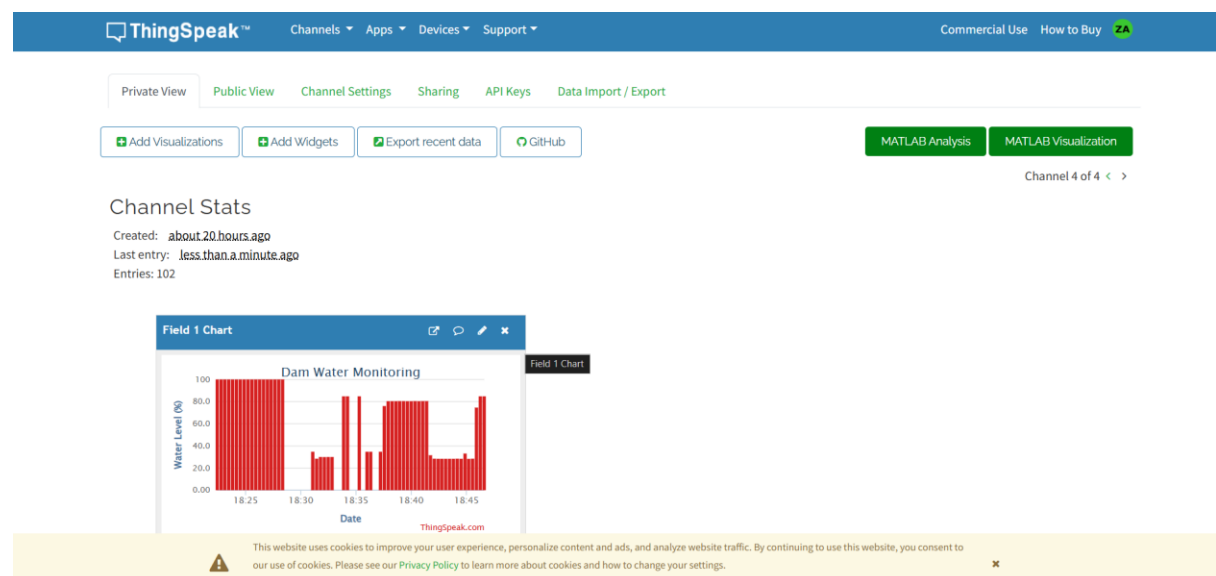
The ESP8266 connects to a Wi-Fi network and periodically uploads water-level data to the ThingSpeak cloud platform. The uploaded data is displayed as graphs and can be monitored remotely through a web browser.

Channel ID: 3403801

7. Results

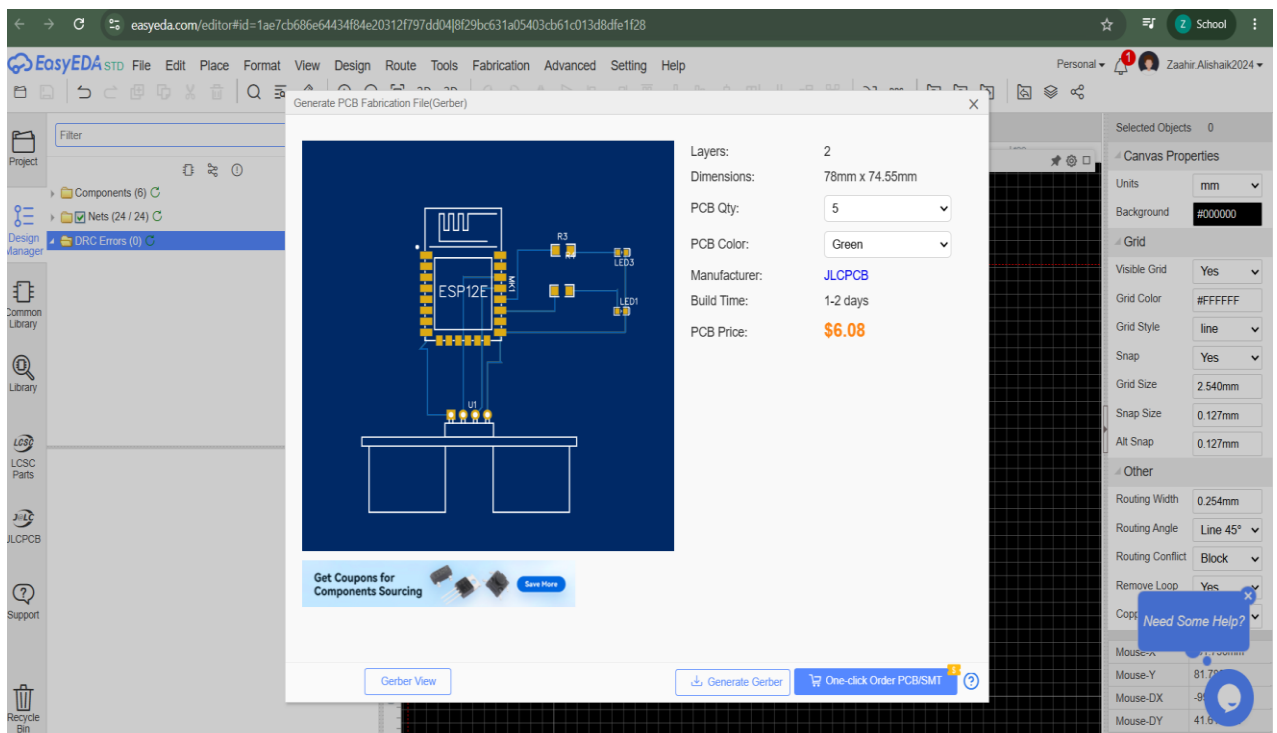
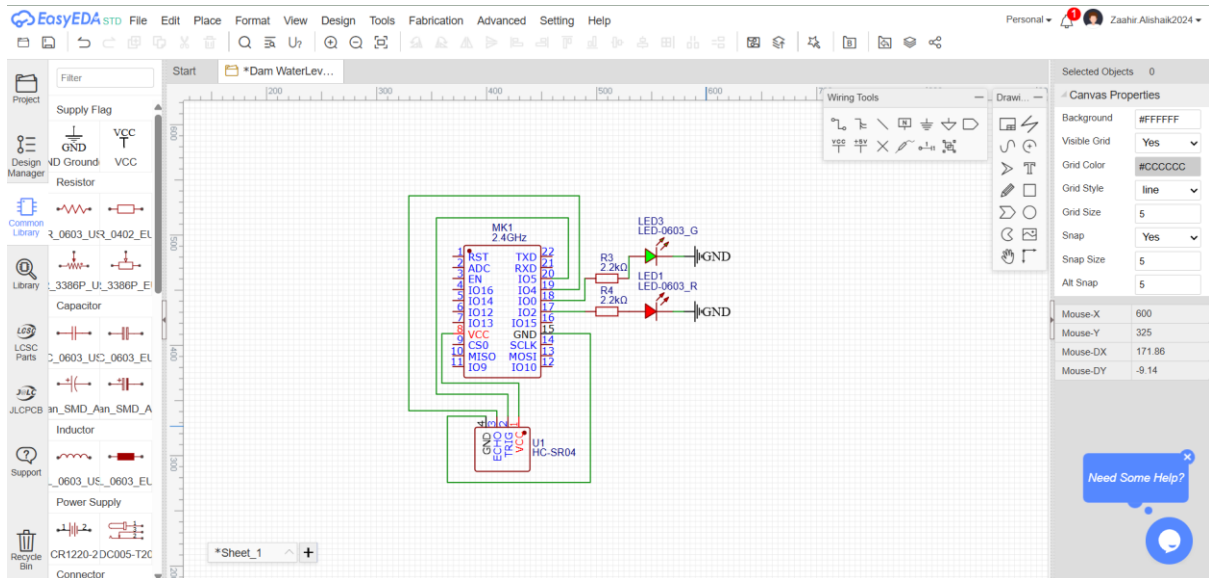
The developed prototype successfully:

- Measured water levels using the HC-SR04 sensor.
- Calculated water level percentages.
- Generated Safe, Warning, and Danger alerts.
- Uploaded real-time data to ThingSpeak.
- Displayed water-level trends through graphical visualization.



8. Gerber File Generation

A PCB layout was designed using EasyEDA. Design Rule Check (DRC) was performed successfully, and Gerber files were generated for PCB fabrication purposes.



9. Conclusion

An IoT-based Dam Water Level Alert System was successfully designed and implemented using ESP8266 NodeMCU and HC-SR04 ultrasonic sensor. The system provides reliable water-level monitoring, alert generation, and cloud-based data logging through ThingSpeak. The prototype demonstrates a low-cost and effective solution for water-level monitoring applications.